

Tejeshwar C D R

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SUMMARY

Computer Science & Engineering student at Amrita Vishwa Vidyapeetham with a focus on research-driven development. Specializing in Evolutionary Computing, Knowledge Engineering, and Deep Learning, with a specific interest in the architectural design of optimization engines. Dedicated to extracting deep technical insights to develop innovative solutions for the logistics sector and personal productivity. Currently researching Ontology-based knowledge integration within Evolutionary Algorithms to enhance intelligent decision-making and system adaptability.

PROJECTS

CFRL-Enabled Edge-Cloud Sepsis Prediction Framework

[Link to Project](#)

- Developed a hybrid Computational Intelligence framework using LSTMs and Genetic Algorithms to capture nonlinear temporal patterns for early sepsis detection.
- Applied Knowledge Distillation to compress cloud-trained LSTM intelligence into lightweight GRU models, achieving 99.9% performance retention for real-time edge deployment.

Hive Mind – Robotic Swarm Foraging Simulation

[Link to Project](#)

- Engineered a bio-inspired multi-agent system using a Finite-State Machine (FSM) to coordinate autonomous foraging and homing behaviors in e-puck robots.
- Implemented Particle Swarm Optimization (PSO) principles and emitter-receiver communication to enable adaptive memory-driven exploration and cooperative information sharing.

AI-Integrated Rapid Compliance Checker (LogiTHON '25 Finalist)

- Engineered an automated pipeline for cross-border shipment verification, enabling real-time detection of restricted items and destination-specific regulatory conflicts.
- Integrated an AI-driven document generation module to streamline compliance reporting, securing a finalist position at the premier logistics hackathon at IIT Bombay.

Intelligent Resume Management & Adaptive ATS System

- Designed a prototype for a dynamic resume generation system that utilizes AI semantic analysis to map repository-based projects to specific job descriptions.
- Formulated a conceptual framework for a personalized ATS scoring system and keyword optimization engine to enhance resume rectification during initial screening.

EDUCATION

2023 – 2027	B.Tech, Computer Science and Engineering Amrita Vishwa Vidyapeetham, Coimbatore	<i>CGPA: 7.71/10</i>
2022 – 2023	Central Board of Secondary Education (Class XII) SRV Innovative Senior Secondary School, Rasipuram	<i>Percentage: 85%</i>

PUBLICATIONS

Comparative Analysis of Carbon Footprint Predictions for Electronic Product Categories... | *SoCTA - 2025, NIT Jalandhar*

- Authored a research paper evaluating Random Forest, Federated Learning, and Contrastive Learning approaches for carbon emission prediction.
- Demonstrated that Federated Learning achieves near-centralized accuracy while preserving data privacy across distributed nodes.

SKILLS

Core AI & Research	Evolutionary Computing (GA, PSO), Deep Learning (LSTM, GRU), Fuzzy Logic, Knowledge Engineering.
Semantic Technologies	Ontology-based Knowledge Integration, Semantic Data Modeling, Knowledge Graph Representation, RDF/OWL.
Optimization	Multi-Objective Optimization, Genetic Hyperparameter Tuning, Adaptive Thresholding, Particle Swarm Logic.
Architectures	Edge-Cloud Collaborative Frameworks, Knowledge Distillation (Model Compression), Federated Learning, Agentic Workflows.
Data Engineering	Real-time Physiological Data Processing, Class Imbalance Mitigation, Temporal Sequence Formation, Feature Engineering.
Programming	Python (PyTorch, TensorFlow, Scikit-learn, NumPy), C/C++.
Robotics & Tools	Webots (Robotics Simulator), e-puck Robot Integration, Emitter-Receiver Communication, Finite-State Machines (FSM).
Version Control	Git, GitHub Actions.
Documentation	L ^A T _E X, Technical Reporting.